



SKG14 IC8 Pacing & Emerging Technologies for ICM Engineers,L2

7.0 References, Standards and Further Reading

July 2025

The PETRONAS Group adopts zero tolerance against all forms of bribery and corruption. We abide by the PETRONAS Code of Conduct and Business Ethics (CoBE) & Anti-Bribery and Corruption (ABC) Manual, guided by our Shared Values and Statement of Purpose.

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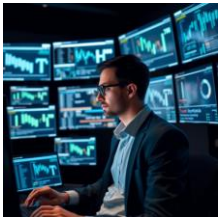
7.1 External Standards

External Standards



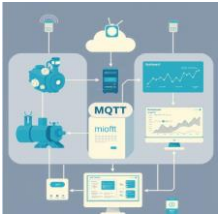
ISO/IEC 30141:2024 - Internet of Things (IoT) - Reference Architecture

1. Provides a standardized framework for designing IIoT systems.
2. Includes principles and concepts for system development and deployment.
3. Addresses functional components like connectivity, security, and data analytics



IEC 62443 Series - Industrial Automation and Control Systems Security

1. Focuses on cybersecurity in industrial environments, a critical aspect of IIoT. But it is not specific to IIOT.
2. Elements on the assessment of components (Sections 4-1 and 4-2) can be used for IIOT security assessments



ISO/IEC 20922:2016 – Information Technology - Message Queuing Telemetry Transport (MQTT)

1. Standard that specifies the protocol
2. Includes security requirements



IEEE P2413 - Standard for an Architectural Framework for the Internet of Things (IoT)

1. Defines a standardized architecture for IoT, including IIoT.
2. Supports interoperability among IoT systems and verticals.



ISO/IEC 3016X Series

Series of standards on IOT covering general requirements, architecture and compatibility requirements

7.2 Internal Standards

Internal Standards



1.PTS 14.11.05: Cyber Security Management for Operational Technology

1. PETRONAS Standard for OT Cyber Security
2. Contains security reference architecture showing how IIOT coexists with OT.

2.PTG 14.11.07: OT Cyber Security Network and System Defence in Depth Guideline

1. Guidance on how to implement PETRONAS approach to cyber security.

3.PTG 14.40.13: Secure by Design for IIOT System

1. Guidelines on developing an IIOT system including reference architecture, risk assessment approach and security baselines.
2. Contains RACI for IIOT projects and operations and standardized naming convention.

4.PTS.F08.NA.55844 : PETRONAS Digital Foundation and ICT Standards (PDFIS)

1. PETRONAS Enterprise Architecture Reference Architecture for IIOT

SUPPLEMENTARY READING

1. MylearningX IIOT Module

